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Decomposition Reaction of Lithium Bis(oxalato)borate in the Rechargeable Lithium-Oxygen Cell

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In the Li-oxygen battery, the reactivity of superoxide radicals with the electrolyte system during discharge is known to be one of the key parameters to determine the nature of discharge product. Here, we report that lithium bis(oxalato)borate, a common electrolyte salt, is readily decomposed by superoxide radicals. The solid discharge product from cells using this electrolyte salt was identified as lithium oxalate through ex-situ X-ray diffraction and infrared spectroscopic observations. The nucleophilic substitution reaction at the boron centre by superoxide radicals, followed by a series of reduction and gas evolution reactions, are proposed to account for the overall process which also has implications for other salts.

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The idea of utilizing oxygen from air as a possible energy source has lead to the development of many metal-air battery systems such as zinc-air, aluminum-air, iron-air, magnesium-air, and lithium-air as well as fuel cell systems.¹⁻⁴ Amongst these, the lithium-air cell possesses one of the highest energy densities ($\sim 5000 \text{ Wh kg}^{-1}$ based on O_2) and a high open circuit potential.⁵ Abraham et al. were the first to employ an aprotic electrolyte system where lithium ions transfer charge, and to identify the reaction product on the cathode side as lithium peroxide.^{6,7} They proposed the overall reaction involved is: $\text{O}_2 + 2\text{Li} \rightarrow \text{Li}_2\text{O}_2$. The standard electrode potential from the change in the standard Gibbs free energy of formation is 2.959 V (where $\Delta G^\circ = -570.954 \text{ kJ mol}^{-1}$).⁸ The porous carbon structure is required to accommodate the insoluble discharge product (Li_2O_2) on the cathodic side as well as to facilitate oxygen diffusion to the reaction site through the cathode film. In addition, this porous carbon network must provide enough network conductivity to deliver electrons to the reaction site smoothly and lower the overall impedance. A homogenous distribution of nano-sized catalyst is also required to maximize the performance.

The mechanistic details concerning the reduction of oxygen on glassy carbon surfaces in a aprotic solvent were previously investigated with tetrabutylammonium (TBA) salts such as TBAPF_6 and TBAClO_4 using cyclic voltammetry and rotating disc electrode techniques.^{9,10} These studies revealed that the initial step of oxygen reduction in an aprotic solution yields superoxide (O_2^-), which is thermodynamically stable when bonded to TBA^+ . It can be reversibly decomposed back to TBA^+ and oxygen through a one electron process. But when superoxide is further reduced to peroxide, the reverse peroxide decomposition reaction to superoxide or oxygen requires a high anodic polarization. When the cation was replaced by lithium ion in similar study, they observed that lithium superoxide is formed initially in a similar manner.^{7,10} However, lithium superoxide (LiO_2) is unstable and readily decomposes to lithium peroxide either by a one-electron electrochemical process or by a chemical dismutation reaction. The reverse reaction that involves the decomposition of lithium peroxide is difficult to reverse. However, they claimed that although the reduction of oxygen under the presence of lithium ion is irreversible, the electrolyte property such as solvent donor number¹¹ influences on the 'degree' of irreversibility by forming quasi-stable Li ion-solvent complex.¹⁰ Many other groups have resorted to the development of an efficient catalyst to promote the reversibility, focusing on the control of morphology and surface defect sites of conventional catalysts adopted from the aqueous metal-air system.¹²⁻¹⁴ The high catalytic surface area derived from the nano-sized catalyst and highly porous electrode structure not only produces a high discharge capacity, but more specifically lowers the charging potential dramatically.

This is of key importance to avoid carbon corrosion and reduce the electrolyte oxidation and will have an enormous impact on the cycling behavior.

One of the most pressing issues for non-aqueous Li-air cells, nonetheless, has turned out to be the stability of the electrolyte system in the presence of superoxide radicals.¹⁵⁻¹⁷ It is well known that the reduction of oxygen in an aprotic solvent creates the highly reactive superoxide anion (radical), O_2^- .^{9,18} Unfortunately, this species attacks many common electrolyte solvents, leading to precipitation of undesirable products.¹⁹ For example, propylene carbonate (PC) is reported to undergo ring opening to form a linear carbonate *via* nucleophilic addition of the superoxide radicals on the CH_2 group proximal to the carbonyl moiety.^{15,16} Further decomposition reactions result in formation of lithium propyl dicarbonate, lithium formate, lithium acetate and lithium carbonate as well as CO_2 and H_2O on discharge.¹⁶ In this paper, we report that the common electrolyte salt, lithium bis(oxalato)borate (LiBOB), is decomposed as a result of its reaction with superoxide radicals, as we communicated previously.²⁰ We propose a mechanism which involves the nucleophilic substitution of superoxide on the boron centre followed by further reduction and gas evolution reactions.

Experimental

Materials synthesis.—The $\alpha\text{-MnO}_2$ nanowire catalyst was prepared by a literature method.²¹ Typically KMnO_4 (0.1264 g) and NH_4Cl (0.0428 g) were dissolved in water (40 mL) and stirred for 1 h at room temperature. The solution was transferred to a TeflonTM lined hydrothermal reactor (capacity: 60 mL). The reactor was maintained at 140°C for 24 h in the oven. Afterwards, the reaction mixture was filtered and washed thoroughly with distilled water. The resulting brown powder was dried at 120°C .

Materials characterization.—Powder X-ray diffraction was performed using a Bruker D8-Advance powder diffractometer using $\text{Cu-K}\alpha_1$ radiation ($\lambda = 1.5405 \text{ \AA}$). Fourier transform - infrared (FT-IR) spectra were measured using a Bruker Vertex 70 FT-IR spectrometer in the spectral range from 400 to 4000 cm^{-1} .

Electrochemical measurements.—SwagelokTM-type cells were assembled using a porous carbon cathode together with a metallic lithium anode and a glass membrane separator. The cathode side was exposed to a pressure-balanced chamber filled with 1 atm oxygen gas, employing 0.5 M LiBOB in PC as the electrolyte. The porous cathode was fabricated following a well developed literature method²²⁻²⁴ using carbon black, catalyst, KynarTM 2801 binder and dibutyl phthalate as a pore forming agent. The charge-discharge cycling studies were carried out between 2.0 V and the desired higher potential cutoff. A constant current of 70 mA g^{-1} (with respect to carbon) was applied to both the charge and discharge process.

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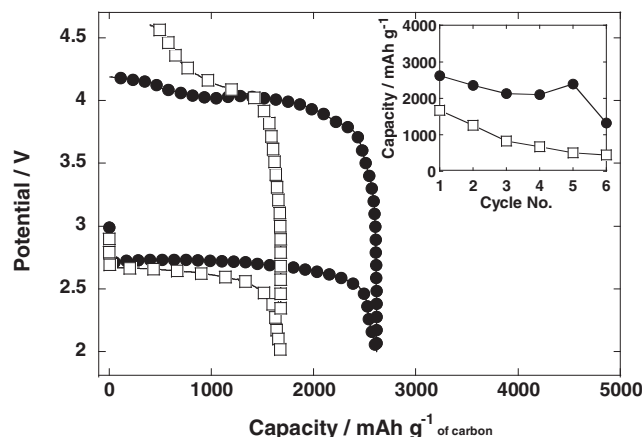


Figure 1. Discharge-charge curves with and without α -MnO₂ catalyst in 0.5 M LiBOB/PC showing the discharge capacity variation with cycling (inset).

Results and Discussion

The Li-oxygen cell performance with an α -MnO₂ nanowire catalyst - reported to be one of the best catalysts for the lithium-O₂ cell¹² to date - was evaluated with 0.5 M LiBOB in PC electrolyte, and compared to that of carbon (Ketjen black) alone. In aqueous solution, amorphous carbon functions well as an oxygen reduction catalyst since many sp³ dangling bonds on the surface contribute to the destabilization of the O=O bonding, thus generating peroxide anions.¹⁸ It is reasonable to assume that amorphous carbon acts as a catalyst for oxygen reduction in aprotic solution in a similar way. Figure 1 shows the 1st discharge-charge profiles and cycling stability (inset) of the Li-oxygen cell with and without the catalyst. The catalyzed cell (~2600 mAh g⁻¹) shows a much higher discharge capacity than the non-catalyzed cell (~1600 mAh g⁻¹). The discharge voltage of the catalyzed cell (2.72 V) is also 60 mV higher than the non-catalyzed cell, which is 2.66 V. These observations imply that oxygen reduction in the presence of metal oxide catalysts proceeds by a slightly different mechanism from carbon alone, and that α -MnO₂ is a more efficient catalyst for oxygen reduction in this electrolyte system than amorphous carbon itself. The theoretical equilibrium potential for formation of Li₂O₂ via oxygen reduction is 2.959 V.^{13,25,26} Thus the overpotential related to oxygen reduction in the presence of the α -MnO₂ catalyst is approximately 0.24 V, which is similar to that observed in an aqueous solution when a highly effective catalyst is used together with a carbon support.¹⁸

In order to determine the discharge product from LiBOB in PC electrolyte, X-ray diffraction patterns and infrared spectra were collected on the discharged electrodes. The catalyzed Li-oxygen cells were discharged to 2.0 V and disassembled inside an argon-filled glove box. The cathode was washed with copious amounts of dry tetrahydrofuran to remove residual electrolyte.⁶ After drying, the sample was loaded in an air-tight holder for X-ray diffraction studies. Figure 2 shows the X-ray diffraction patterns of the discharged electrodes along with a reference pattern that shows the discharge product is almost exclusively composed of lithium oxalate and lithium carbonate. This finding is further confirmed by infrared spectra measured on the same electrode (Figure 3). Though there are some minor contributions (unassigned), the prominent peaks are readily assigned to lithium oxalate,²⁷⁻²⁹ and lithium carbonate. The latter probably arises from PC decomposition *via* superoxide attack.^{15,16} The discharge product using the α -MnO₂ catalyst was also analyzed by the same techniques (Figure 2ii, 3ii). Both X-ray diffraction pattern and infrared spectra indicate that the main discharge product is lithium oxalate and carbonate, although in this case, the FTIR spectra also suggest the existence of lithium triborate. Therefore, irrespective of catalyst, the common discharge products are lithium oxalate (major) and lithium carbonate.

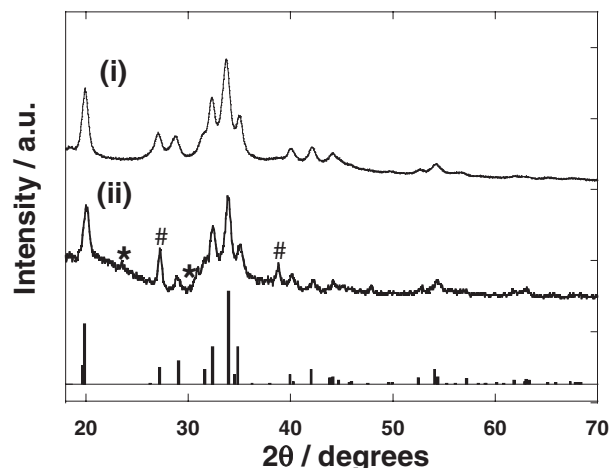


Figure 2. X-ray diffraction patterns of the discharged electrodes (i) without and (ii) with α -MnO₂ catalyst in 0.5 M LiBOB/PC, along with the reference pattern for lithium oxalate (JCPDS 00-024-0646). The symbols, # and * represent reflections from α -MnO₂, and LiB₃O₅, respectively.

The existence of lithium oxalate in the discharged electrodes implies that this material was born from the decomposition of the electrolyte salt, LiBOB.²⁹ Normally, borate, B(OR)₄, is a very stable species since the boron atom is stabilized by adjacent eight electrons from the neighboring four oxygen ligands. However, in the presence of a strong nucleophile (ie., the superoxide radical), chemical reactions such as nucleophilic substitution or addition will ensue. In the system under discussion, decomposition can be readily explained by nucleophilic substitution reaction on the boron centre by the superoxide, and subsequent destabilization of the chelate bond with the neighboring oxygen of the oxalate group in LiBOB (Figure 4). After attack, the reversible nucleophilic substitution reaction is hindered by the fact that the negative charge on the terminal oxygen is greatly stabilized by the adjacent C=O group. This leads to a decrease in the nucleophilicity of the lithium oxalate. Further reaction (a series of

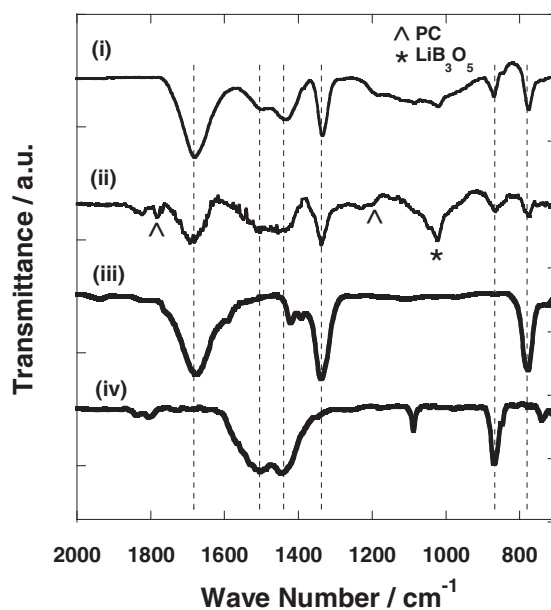


Figure 3. Infrared spectra for discharged electrodes (i) without and (ii) with α -MnO₂ catalyst in 0.5 M LiBOB/PC and the reference spectra for (iii) lithium oxalate and (iv) lithium carbonate.

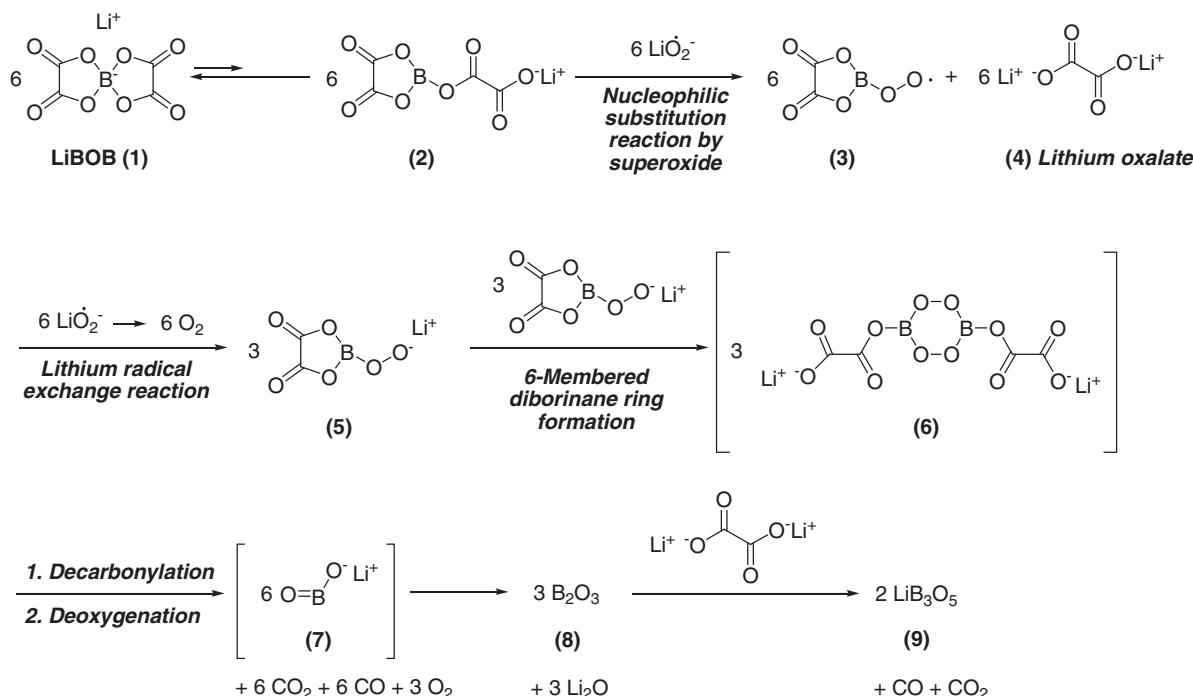
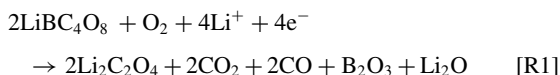
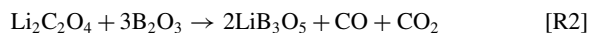


Figure 4. A plausible decomposition mechanism describing the attack on LiBOB by superoxide radicals during the discharge process, and subsequent formation of lithium oxalate (and lithium triborate) as the solid discharge products.

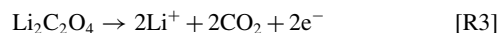
superoxide attack on the boron centre of LiBOB) leads to CO, CO₂ and O₂ gas evolution. The final insoluble products are comprised of lithium oxalate (Li₂C₂O₄), boron trioxide (B₂O₃) and lithium oxide (Li₂O).²⁹ Amongst those decomposition products, boron trioxide and lithium oxide are often amorphous and hence do not appear in the XRD pattern. Because the reduction process involves the formation of lithium oxalate and boron trioxide, the pores in the cathode will be filled with these materials. The pertinent cathodic reaction for the overall process can be written as follows R1.



The existence of lithium triborate (as observed in the FT-IR spectra in case of the α -MnO₂ catalyst) probably suggests that further reaction of boron trioxide with lithium oxalate occurs to some extent. The XRD pattern (Figure 2ii) for the discharged electrode shows a trace of LiB₃O₅, whose formation can be described below by reaction R2:



In Figure 1, the oxidation potential of the Li-O₂ cells in the presence of α -MnO₂ catalysts is observed at 4.01 V, much lower than of the non-catalyzed cells which is poorly efficient on charge. This means that the presence of catalyst indeed plays an important role in lowering the anodic overpotential. However, from the above observations, the charging process actually involves the decomposition of lithium oxalate (and/or lithium carbonate). The appropriate anodic electrochemical reaction for this can be written as follows R3.



For the minor amorphous Li₂O, B₂O₃ or LiB₃O₅ phases, further oxidation is unlikely to occur.

The catalyzed anodic decomposition of lithium oxalate generates CO₂ gas and lithium ions. Figure 1 inset shows the cycling stability of Li-oxygen cells. It demonstrates that the α -MnO₂ nanowire catalyst decomposes lithium oxalate efficiently and gives rise to some cycling stability, whereas the slow decrease in discharge capacity with cycling is probably caused by the corrosion on the anode side and the

accumulation of B₂O₃, Li₂O and LiB₃O₅ phases in the cathode pore structure.

Conclusions

The reactivity of superoxide radicals with the electrolyte system during discharge is one of the crucial factors that determines the nature of the discharge product for rechargeable Li-oxygen cells, and the subsequent ability to recharge the cell. In this work, the stability of the LiBOB/PC electrolyte system was examined. The main solid discharge product was identified as lithium oxalate through X-ray diffraction and infrared spectroscopic studies of the discharged cathode. This reveals that the decomposition of LiBOB occurs as a result of reaction with superoxide radicals. This far exceeds any minor decomposition experienced with LiPF₆ in typical Li-ion cells. The nucleophilic substitution reaction at the boron centre by superoxide radicals followed by subsequent reduction and gas evolution reactions is proposed to account for the overall process. We show the process leads to the precipitation of lithium oxalate and completely hinders formation of Li₂O₂ in the cell irrespective of the electrolyte solvent that is employed. Thus, such cells do not operate as a Li-air battery. This indicates that not only is caution required in selecting suitable electrolyte solvents for the Li-air battery, but for the salts as well. We expect that many salts such as TFSI⁻ (bis(trifluoromethanesulfonyl)imide) and FSI⁻ ((fluoromethanesulfonyl)imide) should be stable to superoxide attack, although reactivity of LiPF₆ bears further investigation.

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